

ES -116

Principles and Applications of Electrical Engineering

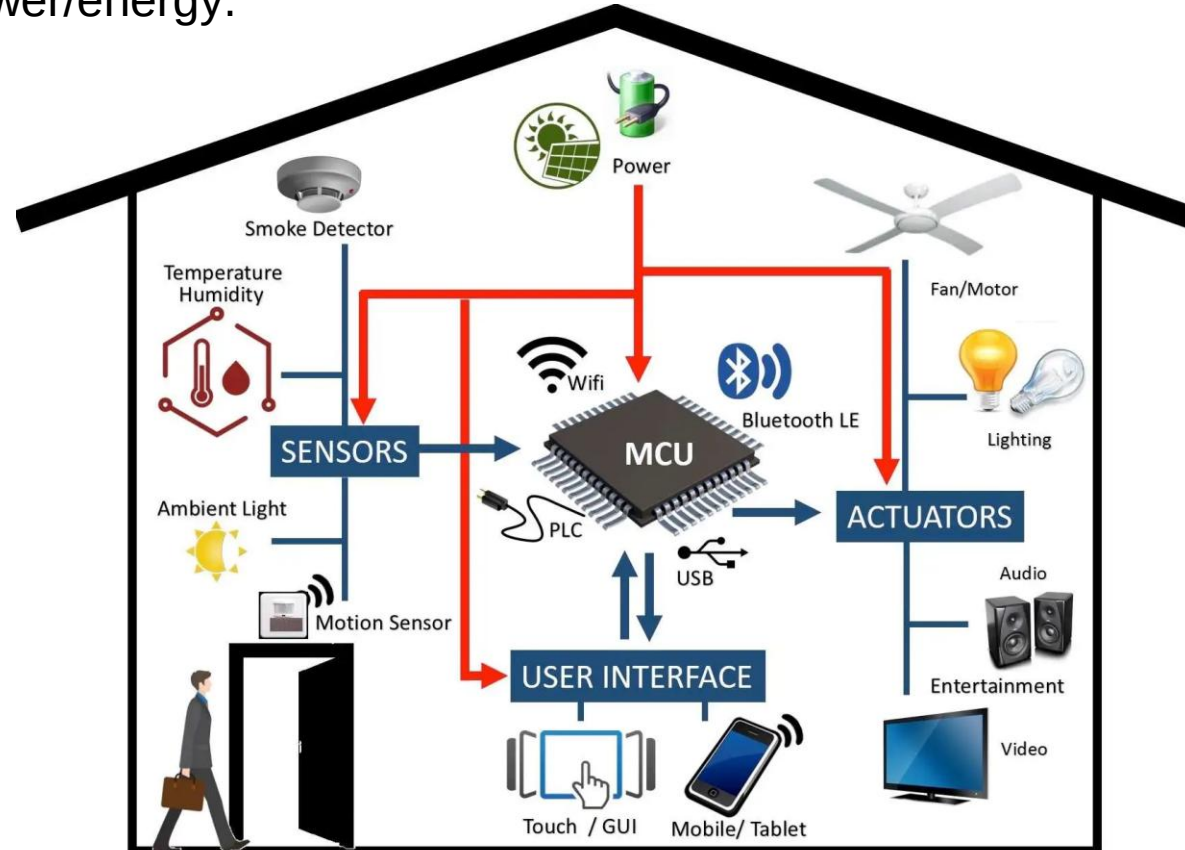
Semester II, 2025-26

How Electricity Became Engineering?

Recap

- To build smart homes and modern electronics, people first had to discover electricity and quantify it in charge/current/voltage/power/energy.

What is Electricity?

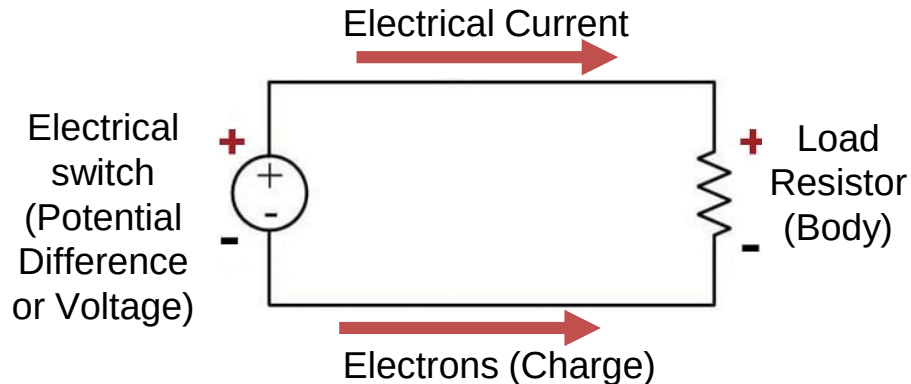


Layman's perspective

- Electricity is an invisible force you suddenly notice when it flows through you from places your hands were never meant to go.

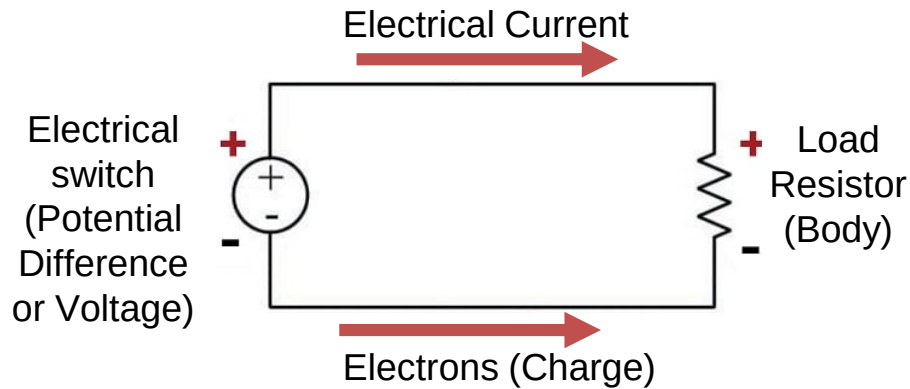
An engineer's interpretation:

- A **large current** flowing through a **resistive load** (your body) when a **potential difference** exists between two points you touch.

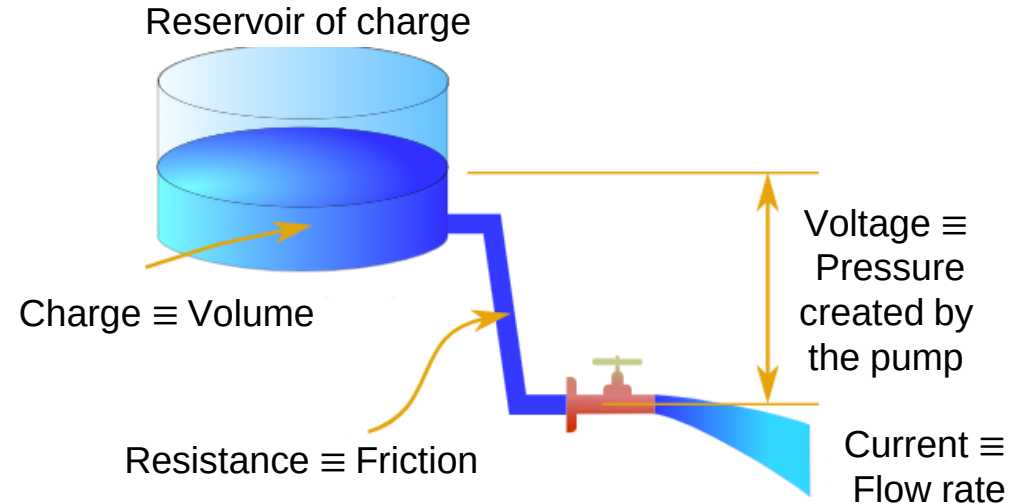


Analogy

- Voltage is like water pressure, current is like water flow, and resistance is how narrow or hard the pipe is for water to pass.

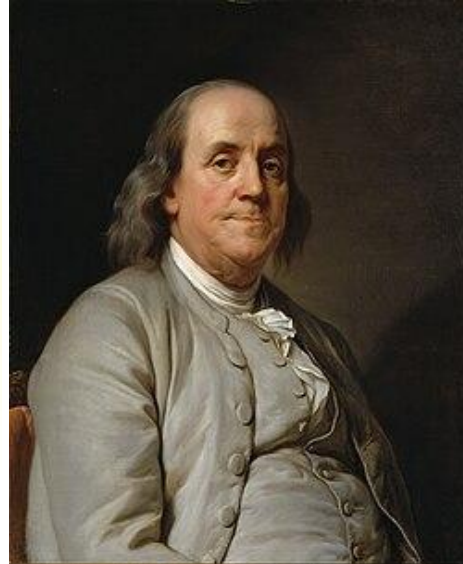
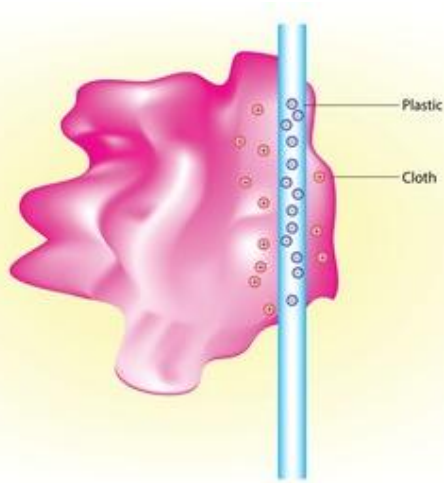


If Q is the charge, current is $I = dQ/dt$.

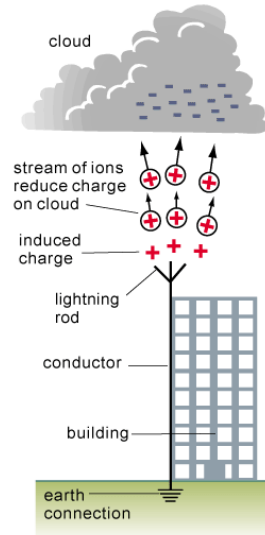


History of Charge

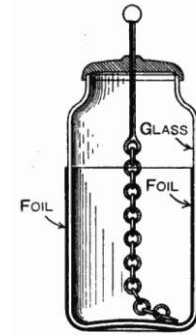
- Charge induction by rubbing a cloth and a insulating rod (Dated before 1600).



- Lightening rod are used to give lightning a low-impedance path to ground, so buildings don't get catch fire when struck.



The kite experiment (Dated 1752)



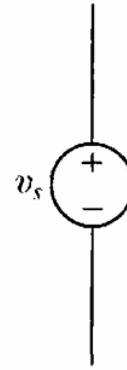
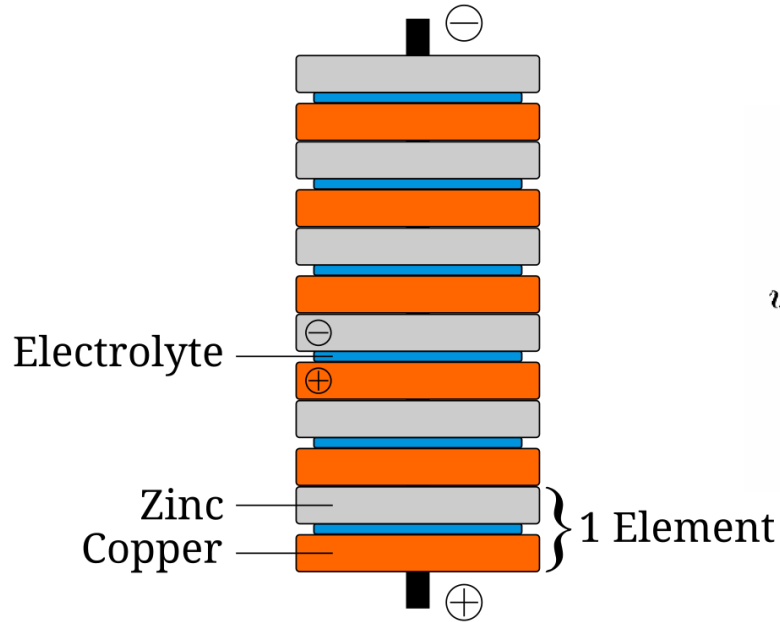
Leyden Jar to store charge

Make the charge flow

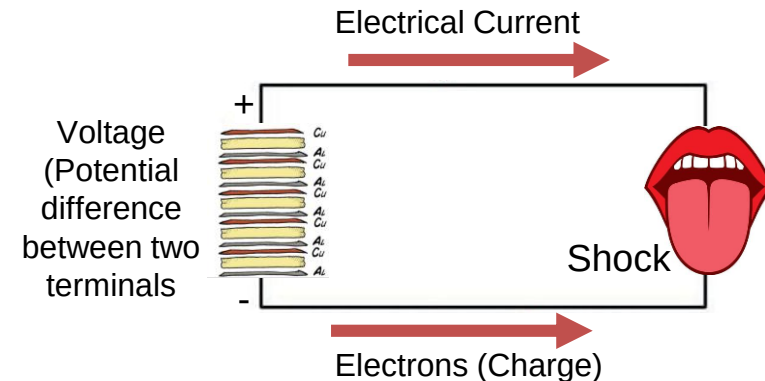
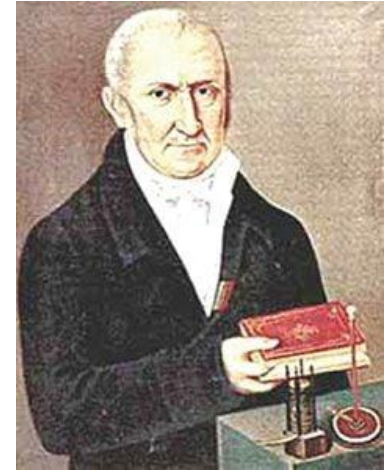
- Invention of a stable battery (voltage source)



Voltaic Pile



- Each copper-zinc pair had a spacer in the middle, made of cardboard or felt soaked in salt water (the electrolyte).
- **An ideal voltage source** is a circuit element that maintains a prescribed voltage across its terminals regardless of the current flowing in those terminals.

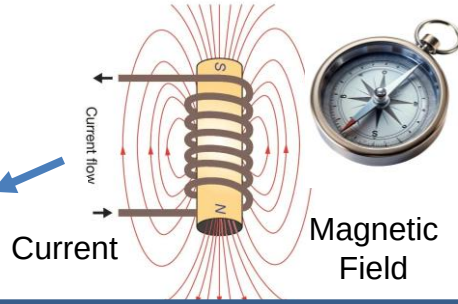


Relating Voltage, Electrical Current and Load

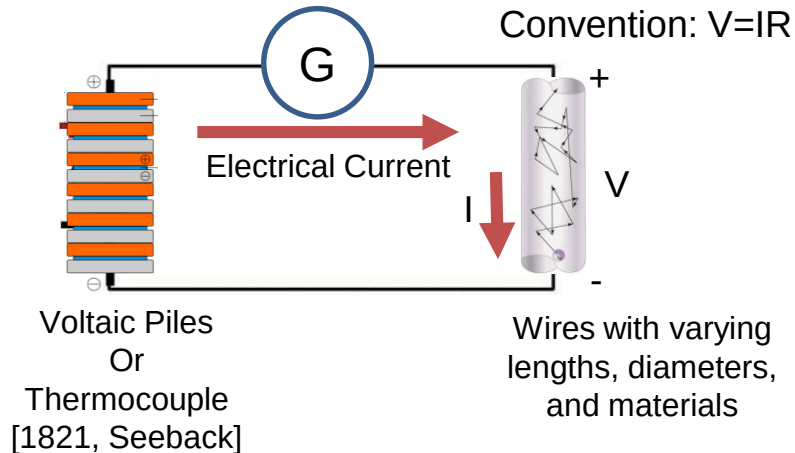
- Galvanometer made the discovery of relating voltage, current to electrical resistance possible.



Galvanometer
[1820, Ørsted]



- Ørsted observed that an electric current flowing through a wire caused a nearby compass needle to deflect. This was the first time a direct link between electricity and magnetism (electromagnetism) was demonstrated.



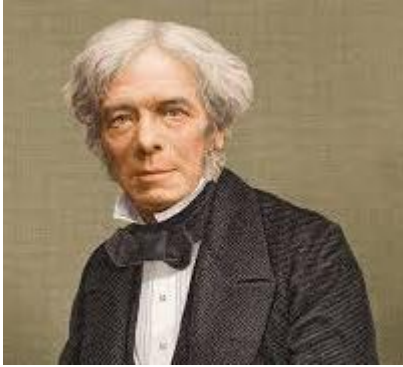
Die galvanische Kette, mathematisch bearbeitet (1827)
The Galvanic Circuit Investigated Mathematically



Observations:

- Current was proportional to voltage and inversely related to resistance by observing consistent V/I ratios for constant wire properties and noting how length/diameter changes affected current.

Question: Is the opposite true?

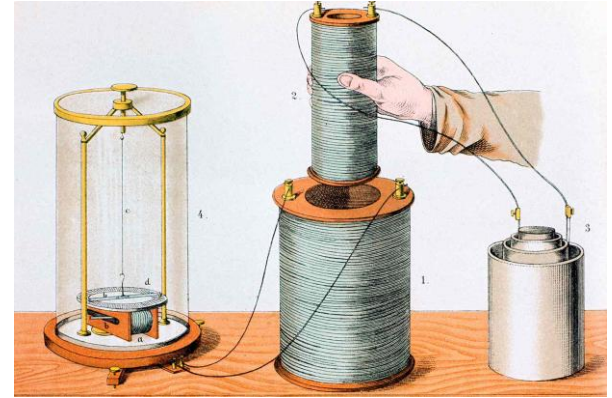


He wondered if the reverse was true: could a magnetic field *produce* an electric current?

First tested if Ørsted was right!



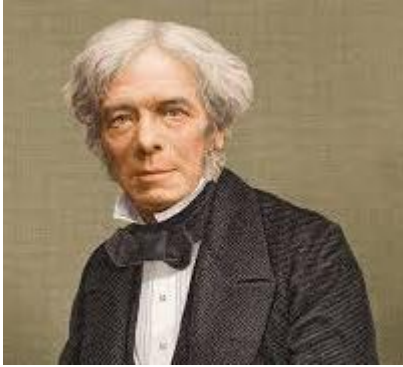
[1821, first electric motor]



Electromagnetic Induction Experiment [1831]

$$\text{(induced voltage) } \varepsilon \propto \frac{d\phi}{dt} \text{ (change in the magnetic flux)}$$

Concept of Dynamo (Faraday's Disc)



He wondered if the reverse was true: could a magnetic field *produce* an electric current?

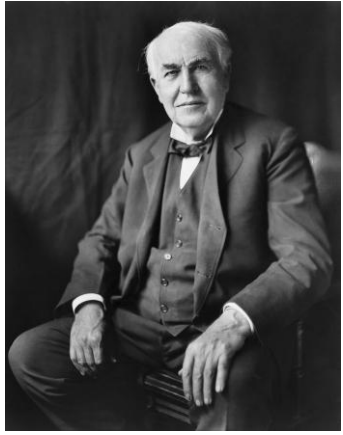
$$\text{(induced voltage) } \varepsilon \propto \frac{d\phi}{dt} \text{ (change in the magnetic flux)}$$

Dynamo (Converting Mechanical to Electrical Energy) [1831]



- A conducting disk spins in a steady magnetic field created by magnets that are perpendicular to the disk.
- Charges in the metal disk move radially due to a Lorentz force exerted by the perpendicular magnetic field.
- It makes one side of the disk (rim/center) becomes at higher potential, creating a DC voltage.
- The generated voltage makes a direct current in an external circuit.

Some decades later...



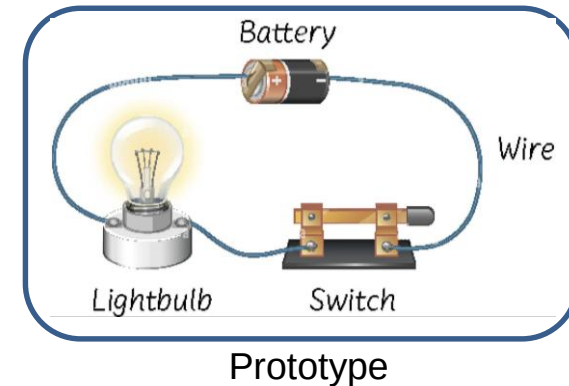
- He used improved DC generators to *power* thousands of light bulbs in real streets and homes.
- Thomas Edison and his team at Menlo Park, West Oran tested thousands of materials.



[1879, Bulb]

Why do we say “Power the bulb?”

- We say “power the bulb” because a bulb is a practical load that uses electrical power for conversion to non-electrical output (produce light).
- There are maximum voltage and currents that can be drawn from the source and minimum voltage/current required by the load.
- Therefore, we define the work done by the source on the load using voltage (V) and current (I) in terms of power, $P = VI$.
- How do you know the time that you battery will last?



Simple Problems

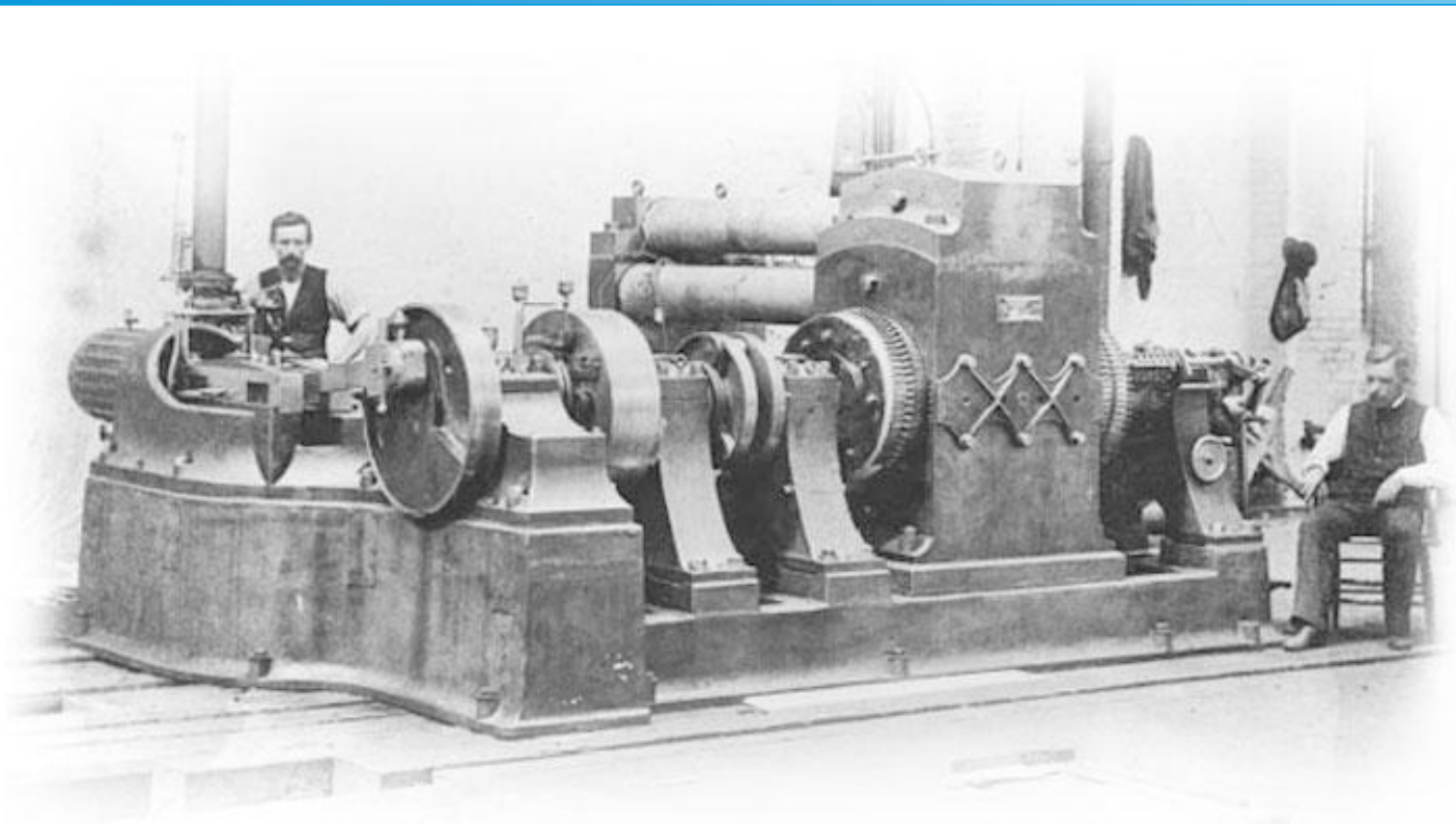
Q. A bulb is connected to a 12 V battery and bulb draws 0.5 A.

- What is the bulb's resistance?
- What is its power?
- How much energy in 2 hours?

Can the given battery with 3.8V, 8.74Wh rating power a 60W bulb with 220-240 voltage rating?



At New York City's Pearl Street Station

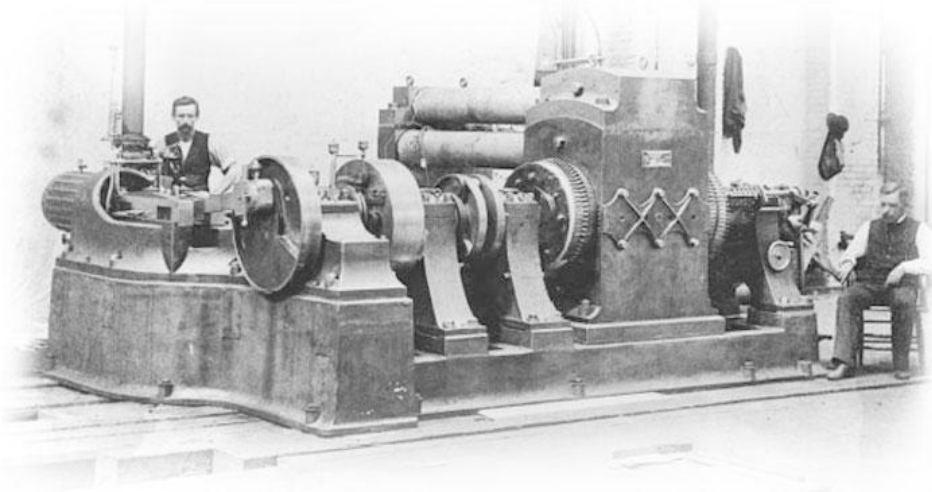


[1882, a large dynamo
“Jumbo” in a large
central power plant]

- Q. If the Jumbo had an output of 100 kilowatts, how many lamps (max. 60 W) can they run?
- Q. How much bill would you charge if they are run for 15 hours? (rate: 10 Rs per kWh)

Analogy

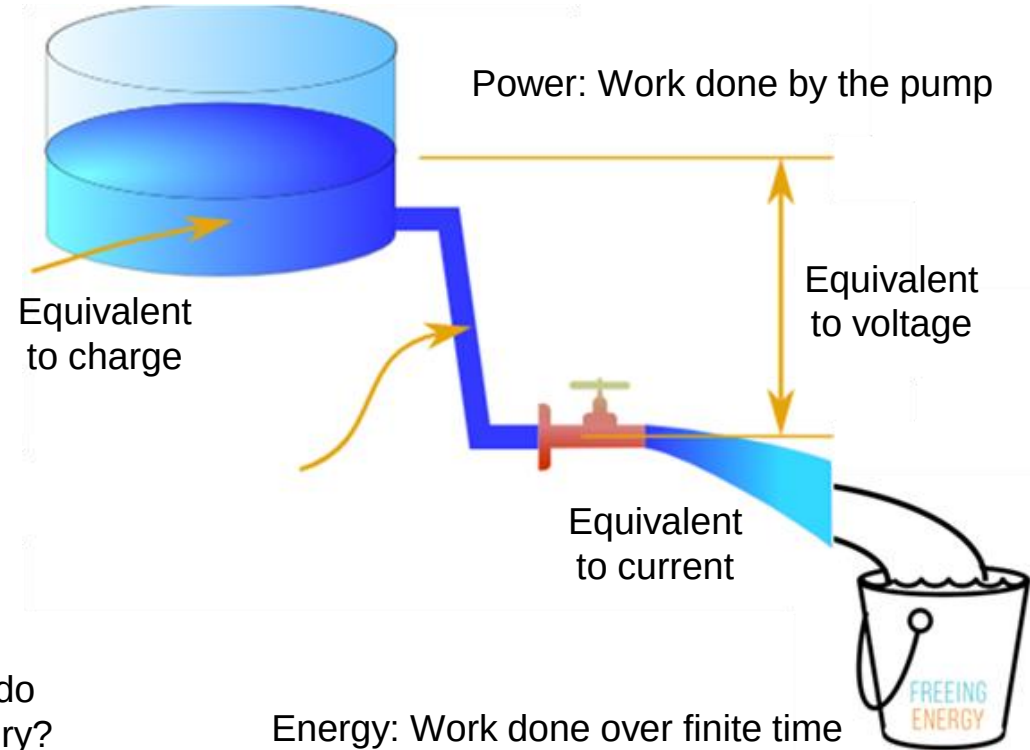
You have energy till the time you have strength!



Finite energy and charge



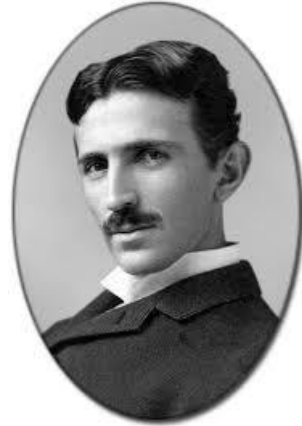
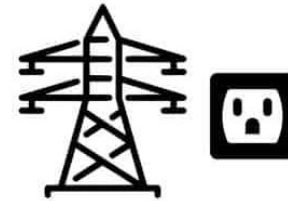
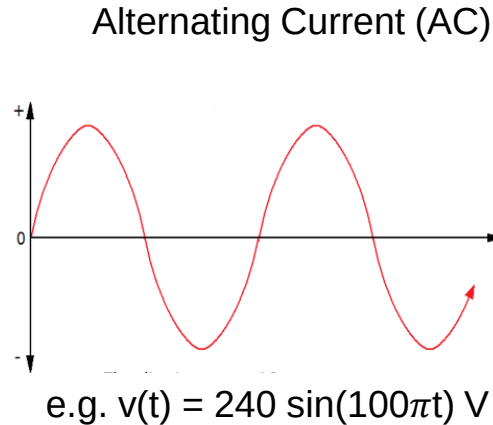
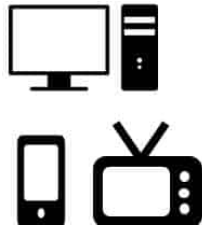
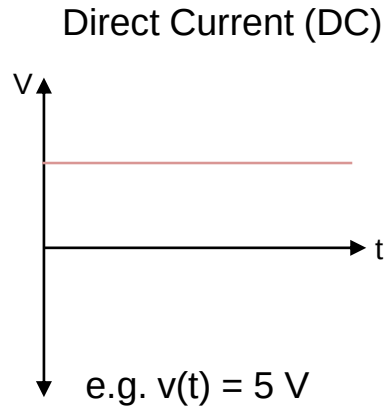
What electrical quantities do you typically see on a battery?



Energy: Work done over finite time

War of currents (AC vs DC)

- But **D**irect **C**urrent (DC) was hard to send far: big losses and many separate stations, high currents.
- Use **A**lternating **C**urrent (AC) that can be transformed up and down in voltage, so we can send power far with small current.



[1888, AC transmission system]

Q. What is the maximum DC and AC signal voltage that can be harmful for human body to touch if 100 mA passing for 2-3 sec can be fatal. (Assume a body resistance of 2 kOhm)

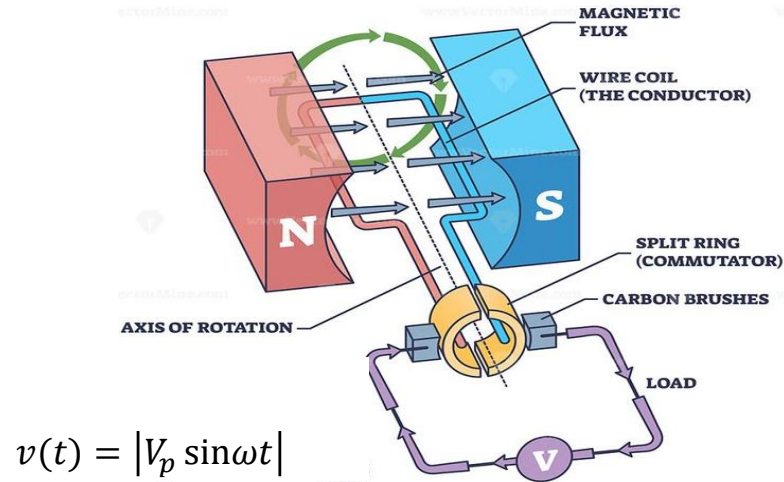
Concept of “Root Mean Square” to compare AC & DC

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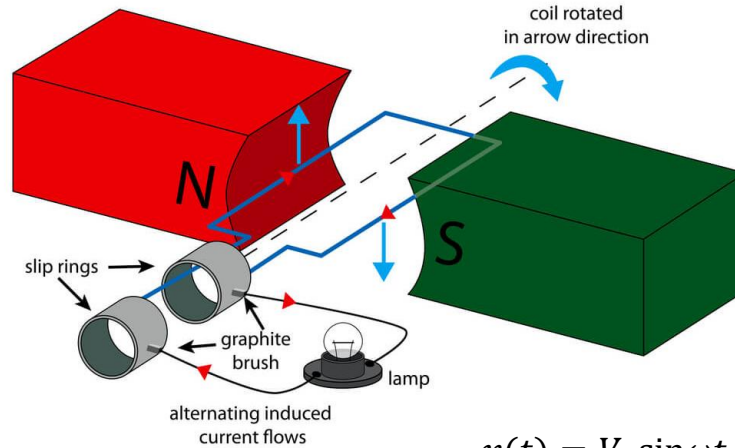
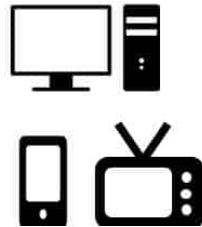
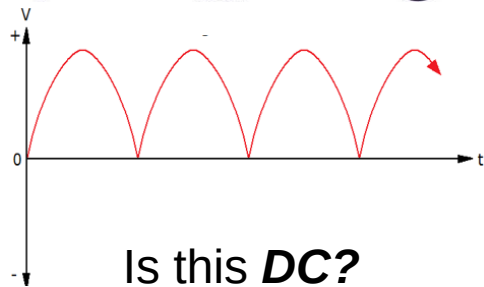
$$v_{DC}(t) = V_{DCmax} \text{ V}$$

$$v_{AC}(t) = V_{AC,p-p} \sin(100\pi t) \text{ V}$$

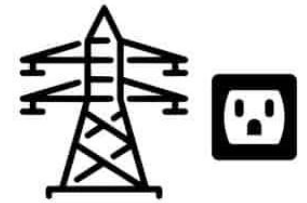
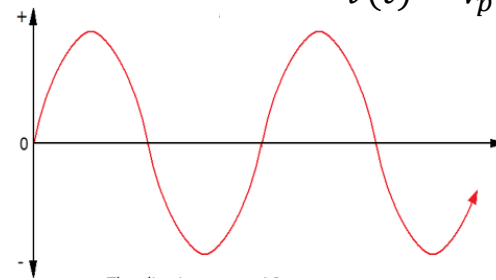
How does actual output of DC/AC generator look like?



$$v(t) = |V_p \sin \omega t|$$

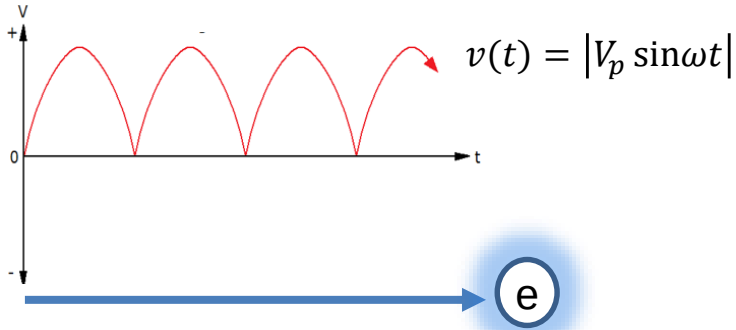


$$v(t) = V_p \sin \omega t$$

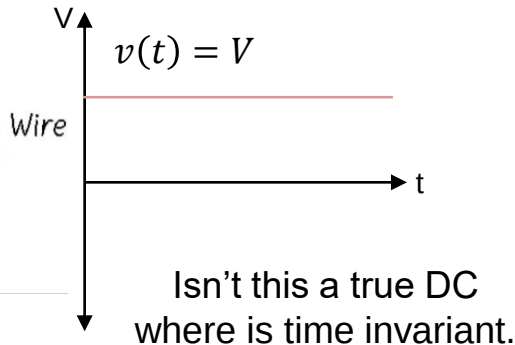
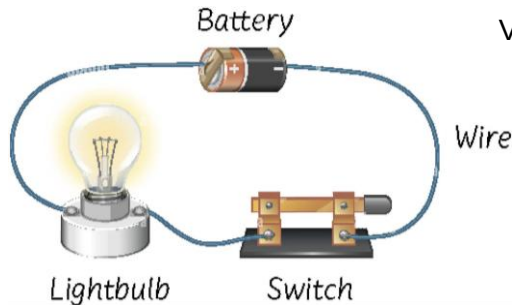


Understanding the difference (AC vs DC)

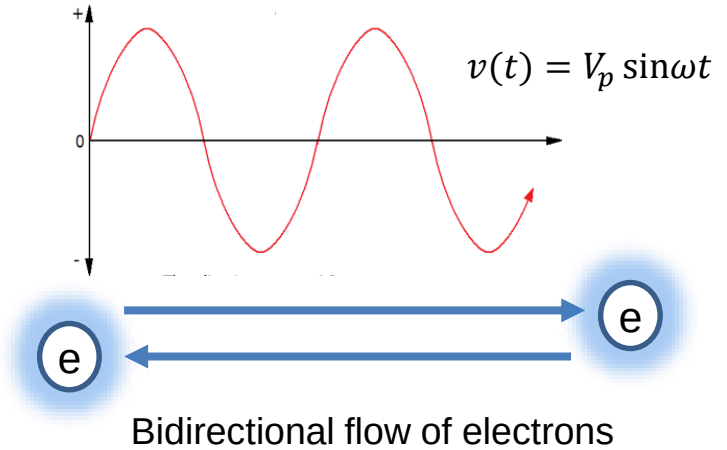
Is this **Direct** current?



Unidirectional, but pulsating flow of electrons



It is **Alternating** as it flows back and forth



Bidirectional flow of electrons

Q. Can you compare **Average** value of three signals?

Average (mean) value of a signal

1. $v(t) = |V_p \sin \omega t|$
2. $v(t) = V_p \sin \omega t$
3. $v(t) = V$

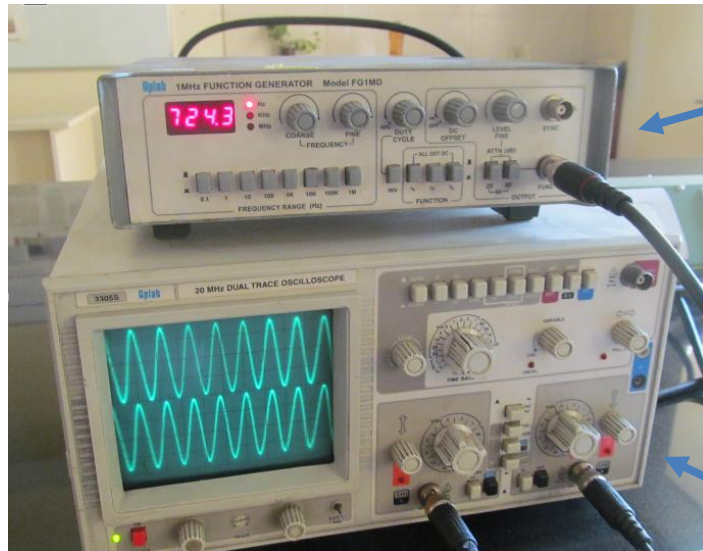
What can you measure practically?

Purely DC signal: Average is non-zero.

Purely AC signal: Average is zero.

→ If the Average (mean value) is non-zero, there is a DC component in the signal.

Lab exercise: Generate different signals from **Function Generators**, Visualize them on **Digital Storage Oscilloscope (DSO)** and Measure different values from **Digital Multi Meter (DMM)**.



Function Generator

Q. What voltage will the multimeter display if you try to read 220V AC from a wall socket?

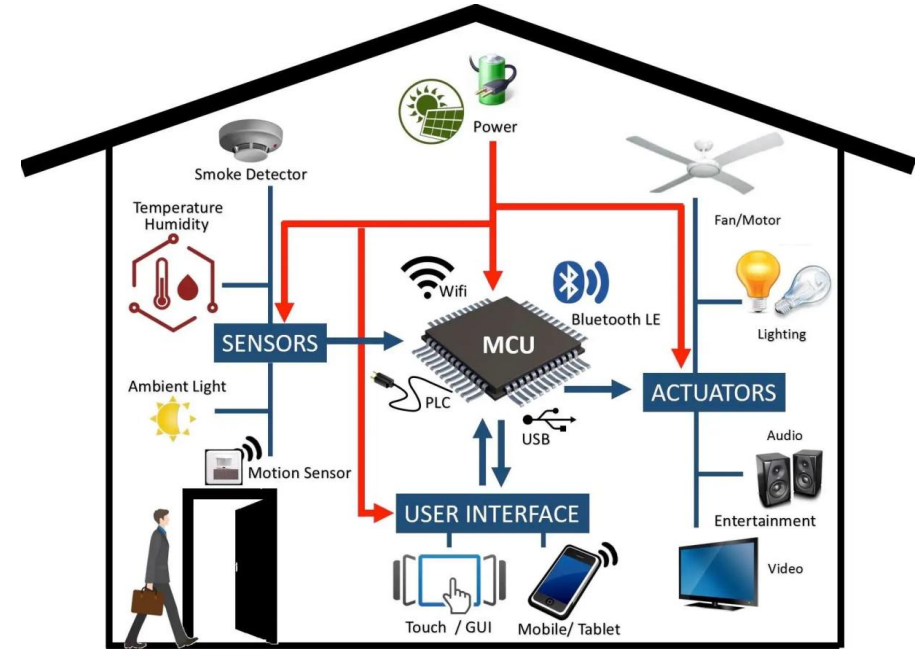
Digital Storage Oscilloscope (DSO)



Digital Multi Meter (DMM)

Summary

- Charge – realizing there is something stored/moved.
- Current – needed for telegraph/lighting: controlled flow in wires.
- Voltage – batteries/generators to push current around homes and cities.
- Power & energy – designing grids, ratings, electricity billing.



- Electrical engineering is about designing useful systems that use and control electricity to do work in the real world.

Practice problem (if time permits)

Q. A 2-wheeler EV uses a battery whose terminal voltage drops linearly from 48 V to 36 V over a 30-minute ride, while the motor draws a constant 20 A during this interval.

- Write $V(t)$ in this interval (take t in seconds, $t=0$ at 48 V).
- Express the instantaneous power $P(t)$.
- Find the total electrical energy delivered by the battery in these 30 minutes.